

Topic 6: Further Mechanics

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the effect of mass, velocity and radius of orbit on centripetal force.

Mathematical skills that could be developed in this topic include translating between degrees and radians and using trigonometric functions.

This topic may be studied using applications that relate to mechanics, for example, transportation.

Students should:	
97.	understand how to use the equation impulse = $F\Delta t = \Delta p$ (Newton's second law of motion)
98.	CORE PRACTICAL 9: Investigate the relationship between the force exerted on an object and its change of momentum.
99.	understand how to apply conservation of linear momentum to problems in two dimensions
100.	CORE PRACTICAL 10: Use ICT to analyse collisions between small spheres, e.g. ball bearings on a table top.
101.	understand how to determine whether a collision is elastic or inelastic
102.	be able to derive and use the equation $E_k = \frac{p^2}{2m}$ for the kinetic energy of a non-relativistic particle
103.	be able to express angular displacement in radians and in degrees, and convert between these units
104.	understand what is meant by <i>angular velocity</i> and be able to use the equations $v = \omega r$ and $T = \frac{2\pi}{\omega}$
105.	be able to use vector diagrams to derive the equations for centripetal acceleration $a = \frac{v^2}{r} = r\omega^2$ and understand how to use these equations
106.	understand that a resultant force (centripetal force) is required to produce and maintain circular motion
107.	be able to use the equations for centripetal force $F = ma = \frac{mv^2}{r} = mr\omega^2$